Transcript: Baryonic Feedback Bias in a DES-Y1-like 3x2pt Likelihood: Unmeasured Because the Pipeline Ignored Baryons

HEP-Genesis-Agent session record
Project: expert-baryon-lensing-bias-3

2026-10-24

The conversation verbatim, generated mechanically from the session record.

A Conversation transcript

The exchange verbatim (assistant text as shown to the user; tool calls summarized by name — see the tool-call log for arguments and results).

User (2026-10-24 19:03:42)

We have two P_hydro/P_grav(k) curves (S(k)) from a HACC baryonic-feedback hydrodynamics emulator, interpolated at k = 0.1, 0.5, 1, 2, 3, 5, 8 h/Mpc, at z = 0, 0.5, 1:

FIDUCIAL point (kappa_w=3.0, e_w=0.6, m_seed=1.3, v_kin=0.65, eps_kin=0.61, omega_m=0.1375, sigma_8=0.8) – standard AGN suppression: z=0: S = [0.999672, 0.991616, 0.976974, 0.948006, 0.915237, 0.856390, 0.802947] z=0.5: S = [0.999774, 0.992759, 0.978298, 0.950904, 0.925438, 0.885998, 0.864809] z=1: S = [0.999888, 0.995517, 0.984937, 0.964613, 0.950974, 0.944283, 0.968580]

WEAK-KINETIC-FEEDBACK point (same but v_kin=0.1, eps_kin=0.02, the minimum of the emulator's design range) – an UNUSUAL regime where S(k) > 1 (power ENHANCEMENT, not suppression) at high k: z=0: S = [1.000136, 1.001737, 1.004678, 1.014715, 1.016499, 0.997683, 0.963204] z=0.5: S = [1.000110, 1.001621, 1.002254, 1.004092, 1.003961, 0.995231, 0.986147] z=1: S = [1.000051, 1.000640, 1.000640, 0.998997, 1.000016, 1.011048, 1.043216]

(k grid for all: [0.1, 0.5, 1, 2, 3, 5, 8] h/Mpc; emulator's native grid spans k=0.022 to 12.56 h/Mpc with 799 points, these are linear interpolations)

I want to know: if a DES/LSST-like cosmic shear 3x2pt analysis assumed gravity-only power (no baryon correction) when the sky actually followed the FIDUCIAL suppression, how biased would the inferred cosmological parameters be? Please:

1. Build a representative 3x2pt-like firecrown likelihood (use firecrown_list_examples to pick a reasonable DES-Y1-like or similar sacc/example setup; standard weak-lensing + photometric systematics is fine, we don't need number counts).

2. Use the CCL baryons mechanism (cosmology.baryons = 'schneider15' or similar, or whatever pk_modifier path is available) to inject something resembling our suppression at z=0 into the theory predictions, OR if that's not directly possible with our tabulated S(k), explain exactly why and what the closest available knob is (e.g. BaryonsSchneider15 parameters

that would approximately reproduce an 8% suppression at k=3 h/Mpc at z=0, 20% at k=8).

3. Compute the theory data vector (firecrown_theory_data_vector) once assuming NO baryon correction (baryons=none) and once WITH your best approximation of our suppression, at the same fiducial cosmology and nuisance parameters.

4. Report how much the chi^2 (firecrown_compute_loglike) changes, and if you can, how much a parameter like sigma_8 or S_8 would need to shift to compensate (even a rough estimate from the augur Fisher matrix is useful – augur_run_fisher or similar).

5. Be explicit about every approximation, every tool failure, and every place where you are not confident in a number. Do not fabricate anything you can't compute from a tool call.

Assistant (quick mode, 2026-10-24 19:03:42)

Tool calls: `firecrown_list_examples`, `describe_desc_tool_family`, `describe_desc_tool_family`, `firecrown_build_likelihood`, `sacc_prepare_for_firecrown`, `firecrown_build_likelihood`, `firecrown_theory_data_vector`, `firecrown_theory_data_vector`, `describe_desc_tool_family`, `firecrown_build_likelihood`, `firecrown_compute_loglike`, `firecrown_scan_loglike`, `namaster_simulat` `namaster_simulate_maps`

I couldn't measure the bias. In this toolset, the firecrown likelihood ignores any baryon setting, so the run "with suppression" gave exactly the same result as the run without it. Below is what did run, where it failed, and what's needed to finish.

What ran

1. Likelihood (partly as you asked).

- I used firecrown's offline DES-Y1-like test file (`firecrown/tests/sacc_data.hdf5`). It is in real space, has 457 points ($\xi+$, ξ-, γ_t and w(θ)) and a full covariance.
- Systematics: shear calibration and photo-z shift per source bin, plus a standard intrinsic-alignment model (NLA). The nuisance values are the DES-Y1 example values (m = 0.012, the listed Δz, A_IA = 0.5, galaxy biases 1.4–2.0). Cosmology: Ω_c 0.25, Ω_b 0.05, h 0.67, n_s 0.96, σ8 0.81, with CAMB and halofit, and no scale cuts.
- **Failure:** you didn't need number counts, so I made a shear-only copy ($\xi+/\xi$-, 227 points). firecrown rejected it: *"Tracer lens0 does not have data points associated with it. Inconsistent SACC object."* The file-preparation tool removes the clustering data but keeps the lens tracers. So I fell back to the full 3x2pt. Removing the lens tracers would fix this, but no tool here does it.

2–3. Two theory data vectors at the same parameters.

- Without baryons: χ^2 against the file's own data = 551.57 (Fig. 6).
- With `baryons = schneider15` (log10Mc = 14.095, η_b = 0.213, k_s = 1000): χ^2 = **551.57, identical**. The tool warned: *"firecrown (pure-CCL mode) uses halofit without baryons/MG here; the cosmology's matter_power_spectrum/baryons/MG switches are ignored."* The two vectors are the same, so this is not a baryon measurement.

4. Partial results that did work.

- **Self-consistency check:** I rebuilt the likelihood on the noiseless no-baryon vector. It gives $\chi^2 = 0.00$ at the fiducial point, so the pipeline recovers its own input.
- **σ8 sensitivity:** with all other parameters fixed, χ^2 changes by 1 for $\sigma 8 = 0.810 \pm 0.0099$ (range 0.800–0.820; Fig. 14). This uses the DES-Y1 covariance with no scale cuts.

Your other questions

Can your tabulated S(k) go in directly? Not here. There is no way to pass a table of S(k, z) in, and the pyccl tools are not available on this server (`No module named 'tools.ccl'`). Even the built-in Schneider15 option is dropped by firecrown's pure-CCL setup. The forecasting tool (augur) also states it does not produce baryonic P(k). Two side notes:

- Schneider15 has no redshift dependence, so it cannot follow your fiducial S(k). At k = 8 h/Mpc your suppression goes from about 20% at z = 0 to about 3% at z = 1, where it is turning back up (S rises from 0.944 at k = 5 to 0.969 at k = 8).
- It also cannot produce the power enhancement (S > 1) in your weak-kinetic point.

Are those Schneider15 numbers a fit to your curve? No. They are untested guesses, and I couldn't evaluate them. Since the call that used them ignored them, I can't say whether they give the 8% suppression at k = 3 or the 20% at k = 8. Please don't take them as calibrated. Fitting them needs a pyccl call that evaluates the Schneider15 suppression for trial parameters, and none is available here.

How much would σ8 or S8 shift? I can't give a number. That needs the change in the data vector caused by baryons, which no tool here could produce. The ± 0.0099 width only tells you that any baryon-induced shift should be compared against about 0.01 in σ8, before marginalising over other parameters. As a rule of thumb (my reasoning, not a tool result), $\xi+$ scales roughly as $\sigma 8^2$, so a constant fractional amplitude drop ϵ in the shear correlations would pull σ8 down by about $\epsilon/2$. Your suppression depends on scale, though, so the real shift would also feed into Ω_m, the IA amplitude and the shear calibration. I did not run a Fisher matrix with augur, because its data vector also has no baryon option.

What's needed to finish

1. A firecrown setup that applies an S(k, z) correction to the power spectrum. firecrown has a `pk_modifier` example type, but this server's likelihood builder doesn't offer it. Otherwise, a pyccl environment where I can apply your table to P(k, z) and compute the shear correlations directly.
2. With the suppressed data vector in hand, the steps are all available: build the likelihood on it, compute χ^2 of the no-baryon model, and profile σ8 to find the best-fit offset. Running a chain on Ω_c and σ8 with nuisance parameters free would give the shift in S8.
3. For a DES-like analysis, add the standard small-scale cuts on $\xi\pm$. The likelihood builder supports this.

Files

All files are in `/Users/nramachandra/hep-genesis-testing/projects/expert-baryon-lensing-bias-3/runs/261024_190326/`:

- `desy1_3x2pt_38aa48.yaml` — the likelihood on the original test data
- `desy1_3x2pt_38aa48_theory_912d53.hdf5` — the noiseless no-baryon data vector
- `desy1_3x2pt_noiseless_0ccc86.yaml` — the likelihood rebuilt on that vector
- `scan_sigma8_63cd84.csv` / `.png` — the $\sigma 8$ profile (Fig. 14)
- `desy1_3x2pt_38aa48_theory_c7d16c.hdf5` — the "with baryons" vector, which is identical to the no-baryon one; don't use it